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**PHENOLOGICAL CHARACTERIZATION OF DRAGON FRUITS GROWN UNDER
CAPIZ STATE UNIVERSITY-PONTEVEDRA CONDITION**

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- ☐ Social Science/Education /Humanities & other related fields
☒ Natural/BiologicalSciences
☐ Food Research/
☐ Developmental/ Computer/Engineering/ Architecture
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- ☐ SSITRDC
 Social Science and Industrial Technology
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☐ LRDC
 Livestock Research Development Center

☐ FFRDC
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(9A) Abstract

A night-blooming, hemi epiphytic climbing cactus called dragon fruit is well-known for its outstanding fruit quality and market potential. Despite the dragon fruit's importance for the economy and the environment, little is known about its phenological development. The reason for conducting this study was the dearth of knowledge regarding the phenological growth stages of dragon fruit, including bud development, shoot development, vegetatively propagated organ development, reproductive development, flowering, fruit development, and fruit maturation. It also describes the cladode characteristics of dragon fruits and how the seasons and climatic factors influence these distinct characteristics. The properties of its fruit were also noted.

The months of the year and climatological information were treated as independent variables in this field experiment, which was set up using a randomized complete block design (RCBD) with four replications. The study was conducted over the course of a year, from May 2021 to April 2023. The results of the study showed that the length of areoles, the width of cladodes, and the number of spines were all strongly influenced by the months of the year. According to fruit features, it was discovered that in dragon, under specific agroclimatic conditions, flowering begins in July and August and ends either in August or September. Likewise, it took an average of 31 days from flowering to harvest. The fruits had an average of 26.73 bracts, a longitudinal diameter of 24.98 cm, a fruit length of 11.11 cm, a weight of 281.36 g for the whole fruit, 206.64 g for the pulp, 73.44% for the pulp yield, 75.55 g for the skin, and 3 mm for the skin thickness. The width of the cladode and the number of spines were influenced by monthly rainfall. The daily temperature and solar radiation had no impact on the cladode characteristics. The daily relative humidity had an impact on the width of the cladode, whereas the dew point had an impact on the length of the areoles, the width of the cladode, the distance between areoles, and the number of spines. It is concluded that the climatological factors such as rainfall, relative humidity, and dew point, as well as the months of the year, had an effect on the development of the cladode characteristics of dragon fruits.

(9B) Keywords

Dragon fruit, Phenology, Cladode, Areole

(10) Introduction

Dragon fruit is a night blooming, hemi epiphytic climbing cactus known for its high fruit value and market potential. Likewise, it is also noted for its rich nutrient contents specially vitamin C, phosphorus, calcium as well as its fiber content, antioxidative and medicinal properties. It also contains betacyanin (water soluble pigments) which impart red color to pulp and peel (Wybraniec et al., 2007).

Like other cacti, dragon fruit is highly adaptable to a new environment due to their ability to tolerate abiotic stress like drought and temperature extremes. Modification of the stem for water storage (succulent), absence of leaves, presence of waxy layer on stem, night-time opening of the stomata and presence of CAM (Crassulacean acid metabolism) photosynthetic pathway enable the plants to adapt under adverse climatic conditions (Nie et al., 2015).

Notwithstanding the economic and ecological significance of dragon fruit, its phenological stages are not well understood. The systematic documentation of phenological stages is essential to achieve higher productivity and better fruit quality since important cultivation practices are based on phenophases of plants. In the words of Kishore (2016), describing all the phenophases of dragon fruit is useful for the adoption of effective crop management practices like nutrient management, pollination, plant propagation, timely harvest of fruits and pest management.

Objectives of the Study

The present study will define the phonological growth stages of dragon fruit, namely: bud development, shoot development, vegetatively propagated organ development, reproductive development, flowering, fruit development and fruit maturation. This work will contribute to the quantitative and qualitative analysis of growth cycle and efficient implementation of crop management practice in the dragon fruit growing in different agroclimatic conditions and in the country as well.

The main purpose of this study was to document and characterize the phonological growth stages of dragon fruits grown under Capiz State University-Pontevedra condition. Likewise, the study would like to:

1. Document the growth stages of dragon fruits grown under Capiz State University-Pontevedra condition in terms of: vegetative bud development, shoot development, development of vegetatively propagated part, reproductive development, flowering, fruit development and fruit maturation;
2. Evaluate the cladode traits in terms of: length of the stem, length of areoles (mm), cladode width (mm), distance between areoles (mm), and number of spines;
3. Characterize the fruits in terms of: onset of flowering, harvesting of fruits, days from flowering to harvesting, number of bracts, longitudinal diameter, fruit length, weight of the whole fruit, pulp weight, skin weight; pulp yield (g plant^{-1}), and skin thickness; and
4. Describe the effect of climate on cladode traits of dragon fruits in terms of stem length, length of areoles, width of the cladode, distance between areoles, and number of spines.

Time and Place of the Study

The study was conducted at the existing dragon fruit plantation of Capiz State University-Pontevedra, Capiz from May 2021 to April 2022.

Scope and Limitations of the Study

The study was limited to the growth stages of dragon fruits in terms of bud development, shoot development, development of vegetatively propagated parts, reproductive development, flowering, fruit development and fruit maturation. Likewise, selective quantitative and qualitative measurements of cladode and fruit characters were also established. The gathered data was analyzed using the descriptive and inferential statistics

(11) Materials and Methods or Methodology

MATERIALS AND METHODS

Materials

The materials needed were as follows: digital weighing scale, tape measure, Vernier caliper, dragon fruit plants, pruning shear, sprayer, humigreen humic acid, sharp knife.

Methods

Experimental Methods

Experimental design and lay-out. This field experiment was laid out in a Randomized Complete Block Design (RCBD) with four replications. The pre-existing dragon fruit plants consisting of 12 rows were divided into 3 rows each. The first three rows, the 4th to 6th rows, 7th to 9th rows, and 10th to 12th rows represented the first, second, third and fourth replications, respectively. The data on the phenological growth stages of dragon fruits were taken from these rows including the cladode and fruiting characteristics.

Selection and marking of experimental plants. Ten randomly selected posts consisting of four dragon fruit plants each from each block were chosen. The sample plants from each post were properly tagged or marked for proper identification.

Construction of canals. To avoid stagnant water in the area during inclement weather condition, canal was constructed in between rows of growing dragon fruit plants.

Cultural Practices

Weeding and cultivation. Weeds were controlled using hand pulling to minimize competition for space, nutrients, light and other resources necessary for better growth of the plants. This was done twice in a month. On the other hand, cultivation of the growing dragon fruit plants was done with the use of shovel to loosen the base of the plant, promote better root aeration and control the growth of weeds. This was done once every six months.

Application of fertilizer. Humigreen with a dilution rate of 100 g per 200 L of water was prepared and applied to the dragon fruit plants once a week.

Control of pest and diseases. In the event that there is an outbreak of pest and diseases, appropriate insecticides/fungicides were applied if it is needed based on the manufacturer's recommended rate. Likewise, infected plant parts were cut and burned to control further spread of diseases.

Harvesting. Fruits were harvested at 34 days of anthesis or when the fruit skin was predominantly reddish.

Data-gathering

For Cladode Traits. The following parameters were gathered for cladode traits such as: stem length, length of areoles (mm), cladode width (mm), distance between areoles (mm), and number of spines.

Length of areoles, cladode width and distance between areoles. Measurement of these parameters was done with the use of a tape measure.

Number of spines. Counting the number of spines was done.

For Fruit Characters. At day 28 of anthesis, the fruits were collected every two days, until 42 days of anthesis. The fruits collected were washed with neutral detergent and water, dried with a paper towel and, subsequently, the following measurements were taken:

Onset of flowering. This was done by recording the month when the dragon plants started to bloom.

Harvesting of fruits. This was done by recording the month when the fruits of the dragon plant in the experimental site were harvested.

Days from flowering to harvesting. This was reckoned from the time when the plants started to bear flowers until such time that the fruits were picked or harvested.

Number of bracts. This was determined by counting the foliaceous scales attached to the surface of the fruits at harvest.

Longitudinal diameter. This was determined by measuring the distance from the base of the fruit to the apex in a circular fashion with the use of a tape measure.

Fruit length. This was determined by measuring the distance running from the base of the fruit to the apex with the use of a tape measure.

Weight of the whole fruit and skin (exocarp + mesocarp). The weight of fruits and skin was determined using a semi-analytical scale, with the parts manually separated and values expressed in grams.

Pulp weight. This was obtained from the difference between total fruit weight and skin weight.

Pulp yield (%). Pulp yield was calculated by the formula: $\text{pulp weight} \times 100 / \text{weight of the whole fruit}$, expressed as percentage.

Skin weight. This was obtained by weighing the exocarp obtained from the samples with the use of a digital weighing scale.

Skin thickness. This was measured using a digital caliper and expressed in mm.

Climatological data. The climatological data were obtained from DOST-PAG-ASA in Roxas City from May 2021 to April 2022 which represented the whole duration of the study: rainfall (mm), solar radiation, temperature, humidity, and dew point.

Statistical Tools and Analysis

Collected data were subjected to appropriate statistical tests.

(12) Results and Discussions

RESULTS AND DISCUSSION

Cladode Traits

Table 1 displays the cladode characteristics unique to dragon plants, including the number of spines, the distance between areoles, the length of the stem, the width of the cladodes, the length of the areoles, and the number of spines. The findings demonstrated that cladode features, such as areole length, areole width, and spine number, were influenced by the months of the year, but not by stem length or distance between areoles.

With means ranging from 37.61 cm to 39.21 cm, it was discovered that the length of the stem of dragon plants was unaffected by the months of the year. The data revealed that the length of areoles varied depending on the month of the year, with mean areole lengths of 0.2975 mm in February, March, and April being the longest, but being comparable to mean areole lengths of 0.2795 mm to 0.29375 mm in September, October, November, and December 2021 and in January 2022. On the other hand, the lengths of plant areoles in May 2021 were the shortest, with a mean of 0.275 mm, although they were comparable to those of plants in June and July 2021, with means ranging from 0.2745 mm to 0.277 mm. The findings show that the length of plant areoles was constant from May 2021 to January 2022, with averages varying from 0.2745 mm to 0.29375 mm.

According to the data, the cladode width of plants was significantly influenced by the months of the year, being the widest in February, March, and April with means of 5.41 cm, 5.41 cm, and 5.43 cm, respectively, but comparable to those plants from October 2021 to January 2022 with means ranging from 5.28 cm to 5.37 cm. Additionally, the data showed that the cladode width was similar to the plant cladode width in July and August, with means ranging from 5.15 cm to 5.22 cm. The cladode width was the smallest in May and June, with means of 5.06 cm and 5.08 cm, respectively. A close examination of the data also showed that the plants' cladode width, with averages ranging from 5.15 cm to 5.32 cm, was comparable to that of those plants from July to November.

Months of the year did not affect the distance between areoles, which ranged from 2.85625 cm to 2.99775 cm. Finally, the months of the year affected the data on the number of spines. Plants in April 2022 had the most spines, with a mean of 3.95, but they were comparable to those in November and December 2021, January 2022, and March 2022, with means ranging from 3.76 to 3.89. The results also showed that plants produced the fewest spines between 3.48 and 3.68 on average during May, June, July, August, September, and October of 2021. Further analysis revealed that plants produced the same number of spines in January, February, and March of 2022 as they did in October, November, and December of 2021, with means ranging from 3.68 to 3.89.

The findings show that months have an effect on the length of areoles, the width of the cladode, and the number of spines, but not on the stem length or the spacing between areoles. Or, to put it another way, the number of spines, the width of the cladode, and the length of the areoles are all determined by the months of the year.

The results of the study's cladode width analysis fall within the range of 5 to 7 cm reported by Rodriguez (1993). The areole's length, however, is less than the 2 to 3 cm finding of the study cited by Donadio (2009). The obtained values for the distance between areoles were somewhat close to Donadio's (2009) finding of 3 to 5 cm. Additionally, the findings of this study support Donadio's (2009) research on the presence of spines, which ranged from 3 to 6.

Table 1. Stem length, length of areoles, width of cladode, distance between areoles, and number of spines as affected by months of the year (May 2021 to April 2022).

Month	Cladode Traits				
	Stem Length (cm)	Length of Areoles (mm)	Width of Cladode (cm)	Distance between Areoles (cm)	Number of Spines
May 2021	37.71	0.2745 ^b	5.06 ^d	2.85625	3.48 ^d
June	37.61	0.2745 ^b	5.08 ^d	2.86	3.48 ^d
July	38.74	0.2745 ^b	5.15 ^{cd}	2.88825	3.48 ^d
August	38.82	0.277 ^b	5.22 ^{bcd}	2.89025	3.52 ^d
September	38.65	0.2795 ^{ab}	5.28 ^{abc}	2.9175	3.61 ^{cd}
October	38.76	0.2845 ^{ab}	5.30 ^{abc}	2.9265	3.68 ^{bcd}
November	38.80	0.29175 ^{ab}	5.32 ^{abc}	2.90875	3.76 ^{abc}
December	38.95	0.29375 ^{ab}	5.35 ^{ab}	2.92275	3.85 ^{ab}
January 2022	39.17	0.29375 ^{ab}	5.37 ^{ab}	2.92725	3.88 ^{ab}
February	39.15	0.2975 ^a	5.41 ^a	2.9325	3.89 ^{ab}
March	39.21	0.2975 ^a	5.41 ^a	2.933	3.89 ^{ab}
April	39.02	0.2975 ^a	5.43 ^a	2.99775	3.95 ^a
P-VALUE	0.540ns	.000**	.000**	0.1893ns	.000**
CV(%)	2.84	2.85	1.29	2.12	2.29

Means with the same letter are not significantly different. ns- not significant, ** - highly significant

Fruit Characters

The parameters used to describe the distinctive characteristics of the dragon fruit grown at the Capiz State University-Pontevedra Campus are presented in Table 2. These included the start of flowering, fruit harvesting, number of bracts, longitudinal diameter, fruit length, weight of the whole fruit, weight of the pulp, yield of the pulp, weight of the skin, and thickness.

In terms of flowering, it was discovered that dragon fruits produced on the Pontevedra campus began to bloom either in July or August of 2021, while other varieties did so in August or September. Fruits were found to be harvested in August or September. Using this information, it was determined that, on average, dragon fruits could be harvested for 60 days for those that began flowering in July and lasted between 59 and 61 days. On average, it took 31 days for plants that began flowering in August—a period that varied from 29 to 33 days—to produce their fruits.

The average number of bracts that dragon fruit produced per fruit was 27, with values ranging from 3 to 34. Fruits' longitudinal diameter, however, ranges from 19 cm to 29 cm, with an average measurement of 24.98 cm. The length of the fruits that were gathered ranged from 8 to 14 cm, or an average of 11.11 cm. The weight of the whole fruits ranged from 114 g to 434 g, with an average weight of 281.36 g. The weight of the pulp varied from 50 g, which was the lightest, to 300 g, which was the heaviest, to an average of 206.64 g. The pulp yield data, which ranges in value from 43.86% to 90.55%, was reported at 73.44% on average.

Skin weight information showed that it ranged from 24 g to 100 g, with an average value of 75.55 g. Finally, it was discovered that skin thickness ranged from 2 mm to 3 mm, with an average value of 3 mm.

The findings show that the fruit properties of dragon fruits grown in CapSU-Pontevedra vary.

The study's results confirm the findings of Castillo-Martinez and Ortiz-Hernández (1994), who discovered that dragon fruit was harvested from May to September. Additionally, the results of Castillo-Martinez and Ortiz-Hernández (1994), Van To et al. (2002), and Martinez (2011), who found that the time needed to harvest dragon fruits after flowering varied from 25 to 41 days is consistent with the above finding. These findings were corroborated by Ortiz and Takahashi (2015), who discovered that pitaya fruits needed to mature between 30 and 32 days after anthesis (DAA).

Although the average fruit weight of 281.36 g is less than the average fruit mass of 442.0 g reported by Osuna Enciso et al. (2007), 437.5 g by Nerd et al. (1999), and 469.2 g by Centurión Yah et al. (2008), the study's results are within the range of 293.1 to 416.2 g reported by Ortiz and Takahashi (2015). The pulp weight result supports the findings of Ortiz and Takahashi's (2015) study, which indicated that it ranges from 87.6 g to 253.30 g, but falls short of the average value of 368.9 g for pulp published by Centurión Yah et al. (2008).

The results are in line with those of Castillo-Martinez et al. (2003), Osuna Enciso et al. (2007), and Castillo-Martinez et al. (2005), who noted that pulp masses for pitaya fruits ranged from 139.6 to 320.1 g and 188.4 to 297.8 g, respectively. Additionally, the findings that pitaya fruits had pericarp masses ranging from 57.9 to 140.6 g and from 72.4 to 120.5 g, respectively, are consistent with Martinez's (2011) and Castillo-Martinez et al.'s (2005) research.

Results of the research conducted by Castillo-Martinez et al. (2005) and Osuna Enciso et al. (2007), *H.undatus* fruits had 73.0% and 79% pulp, which are within the range of pulp yield found in the study, which ranged from 43.86% to 90.55%.

Table 2. Fruit characteristics of dragon fruits at Capiz State University-Pontevedra**campus.**

Fruit Characters	Average	Minimum	Maximum
Onset of flowering		July	August
Harvesting of fruits		August	September
Days from flowering to harvesting (days)	60/31	29/59	33/61
Number of bracts	26.73	3	34
Longitudinal diameter (cm)	24.98	19	29
Fruit length (cm)	11.11	8	14
Weight of the whole fruit (g)	281.36	114	434
Pulp weight (g)	206.64	50	300
Pulp yield (%)	73.44	43.86	90.55
Skin weight (g)	75.55	24	100
Skin thickness (mm)	3	2	4

Effect of Climate on Cladode Traits**Monthly Rainfall and Cladode Traits**

The information in Table 3 illustrates how monthly rainfall affects the cladode characteristics of dragon fruits. With averages ranging from 38.60 cm to 39.19 cm, 0.284 mm to 0.2956 mm, and 2.906 cm to 2.930 cm, respectively, it was discovered that the monthly rainfall had no effect on the length of the stem, the length of the areoles, or the distance between areoles. This indicates that the stem length, areole length, and distance between areoles are constant regardless of whether the monthly rainfall is less than 67 mm, between 67 and 209 mm, or greater than 209 mm.

However, the amount of rainfall every month had an effect on the cladode's length and number of spines. With values of 5.258 cm and 5.261 cm, respectively, the cladode width was greatest when the monthly rainfall was less than 67 mm and lowest when it was between 67 and 209 mm or larger. The number of spines peaked at 3.884 when the monthly rainfall was less than 67 mm and peaked at 3.678 and 3.663 when the monthly rainfall was between 67 and 209 mm, respectively.

The findings show that whereas dragon fruits have the shortest cladode width and number of spines at monthly rainfall above 67 mm, they have the maximum cladode width and number of spines during low rainfall of less than 67 mm.

Table 3. Cladode traits of dragon fruits as affected by monthly rainfall.

Monthly Rainfall (mm)	Cladode Traits				
	Stem Length (cm)	Length of Areoles (mm)	Width of Cladode (cm)	Distance between Areoles (cm)	Number of Spines
<67	39.19	0.2956	5.392 ^a	2.930	3.884 ^a
67-209	38.64	0.285	5.258 ^b	2.906	3.678 ^b
>209	38.60	0.284	5.261 ^b	2.914	3.663 ^b
P-VALUE	0.407^{ns}	.055^{ns}	0.046*	0.695^{ns}	0.017*
CV(%)	2.82	4.17	2.54	2.17	4.98

Means having the same letter superscript are not significantly different. ns – not significant, * - significant at 5% level

Daily Solar Radiation and Cladode Traits

The cladode characteristics of dragon fruits were unaffected by daily solar radiation. The information is displayed in Table 4. The length of the stem, length of the areoles, the width of the cladode, the distance between areoles, and the number of spines were all found to be the same regardless of the amount of daily solar radiation—less than 5 hours, 5-6 hours, and more than 6 hours—with means ranging from 38.471 cm to 38.813 cm, 0.284 mm to 0.289 mm, 5.255 cm to 5.302 cm, 2.904 cm to 2.919 cm, and 3.

This implies that the cladode characteristics of dragon fruits are similarly influenced by daily solar radiation.

Table 4. Cladode traits of dragon fruits as affected by daily solar radiation.

Daily Solar Radiation (hours)	Cladode Traits				
	Stem Length (cm)	Length of Areoles (mm)	Width of Cladode (cm)	Distance between Areoles (cm)	Number of Spines
< 5	38.813	0.284	5.267	2.913	3.673
5-6	38.471	0.2838	5.255	2.904	3.660
> 6	38.716	0.289	5.302	2.919	3.746
P-VALUE	0.675^{ns}	0.426^{ns}	0.596^{ns}	0.818^{ns}	0.381^{ns}
CV(%)	2.86	4.18	2.68	2.17	5.40

Means having the same letter superscript are not significantly different. ns – not significant

Daily Temperature and Cladode Traits

Table 5 lists the effect of daily temperature on the cladode characteristics of dragon fruits. The findings demonstrated that the temperature that existed at the time of the study, which ranged from 21 to 32°C, was the ideal level. The dragon fruit stem measured 38.716 cm in length, with areole lengths of 0.286 mm, cladode widths of 5.281 cm, areole distances of 2.913 cm, and spine counts of 3.706 spines. This suggests that at the time of the investigation, the temperature was at its ideal level, representing the numerous measurements for the cladode characteristics of dragon fruits.

Table 5. Cladode traits of dragon fruits as affected by daily temperature.

Daily Temperature (°C)	Cladode Traits				
	Stem Length (cm)	Length of Areoles (mm)	Width of the Cladode (cm)	Distance between Areoles (cm)	Number of Spines
21-32	38.716	0.286	5.281	2.913	3.706
P-VALUE	-	-	-	-	-
CV(%)	2.81	4.13	2.54	2.17	5.19

Daily Relative Humidity and Cladode Traits

The cladode width of dragon fruits was strongly influenced by the daily relative humidity, but not the stem length, areole length, areole spacing, or spine count. Table 6 displays the outcome.

The results of the statistical analysis of the data revealed that the effects of daily relative humidity on the length of the stem, the length of the areoles, the distance between areoles, and the number of spines in the dragon fruits were similar, with means ranging from 38.671 cm to 39.211 cm, 0.285 mm to 0.298 mm, 2.912 cm to 2.933 cm, and 3.689 to 3.891, respectively.

However, the results showed that the plants' cladode width was largest at a daily relative humidity of less than 60%, measuring 5.414 cm, and lowest at a daily relative humidity of more than 70%, measuring 5.269 cm on average.

The findings show that the daily relative humidity (RH) of the dragon fruit cladodes is influenced similarly by RH levels of less than 60% and more than 70%, with less than 60% RH favoring the growth of larger cladodes.

Table 6. Cladode traits of dragon fruits as affected by daily relative humidity.

Daily Relative Humidity (%)	Cladode Traits				
	Stem Length (cm)	Length of Areoles (mm)	Width of the Cladode (cm)	Distance between Areoles (cm)	Number of Spines
<60	39.211	0.298	5.414 ^a	2.933	3.891
>70	38.671	0.285	5.269 ^b	2.912	3.689
P-VALUE	0.348^{ns}	0.054^{ns}	0.045[*]	0.540^{ns}	0.052^{ns}
CV(%)	2.81	4.13	2.54	2.17	5.19

Means having the same letter superscript are not significantly different. ns – not significant, * - significant at 5% level

Daily Dew Point and Cladode Traits

Table 7 illustrates that the dew point had no effect on the cladode characteristics of dragon fruits, such as the length of the stem, but had an effect on the length of the areoles, the width of the cladode, the distance between areoles, and the number of spines. Regardless of the dew point, the results showed that the stem length of dragon fruits varied from 38.580 cm to 39.002 cm. However, the areole lengths of dragon fruits were longer with a dew point of less than or equal to 250.28, with means of 0.291 mm and 0.298 mm, respectively, as opposed to a dew point of over 250.28, which was 0.280 mm. Similar to this, the plants' cladode width and spine count reached their highest levels when the dew point was lower than or equal to 250.28, with means that varied from 5.246 cm to 5.324 cm and 3.799 cm to 3.948, respectively. Dew point also had an impact on how far apart dragon fruit areoles were, with the maximum distance between cladodes reaching 3.00 cm at a dew point of 250.28 and the shortest distances reaching 2.90 cm and 2.916 cm at dew points greater than or less than 250.28, respectively.

These results show that a dew point of less than or equal to 250.28 enhances the development of the cladode features of dragon fruit, including areole length, cladode width, areole spacing, and the number of spines.

Table 7. Cladode traits of dragon fruits as affected by dew point.

Dew Point	Cladode Traits				
	Stem Length (cm)	Length of Areoles (mm)	Width of Cladode (cm)	Distance between Areoles (cm)	Number of Spines
>250.28	38.580	0.280 ^b	5.211 ^b	2.900 ^b	3.588 ^b
250.28	39.022	0.298 ^a	5.246 ^a	3.000 ^a	3.948 ^a
<250.28	38.818	0.291 ^a	5.324 ^a	2.916 ^b	3.799 ^a
P-VALUE	0.659^{ns}	0.001[*]	0.003[*]	0.014[*]	0.000[*]
CV(%)	2.85	3.81	2.40	2.17	4.18

Means having the same letter superscript are not significantly different. ns – not significant

Climatological Data

Month	Rainfall Climatological Data (mm)	Solar Radiation	Temperature (°C)	Humidity	Dew Point
May 2021	260	8.20	29.27	77.43	338.23
June	174.5	5.81	27.65	76.07	192.466
July	195.50	4.82	29.81	80.33	252.1
August	261.50	7.31	28.65	76	251.77
September	342	5.92	28.17	82.73	252.43
October	195	4.92	28.27	82.23	252.43
November	233.50	6.93	28.68	83.47	255.13
December	184.0	4.36	27.52	81.6	241.9
January 2022	39.5	6.6	26.37	81.17	233.57
February	119.5	5.49	27.25	81.39	245.96
March	71.50	6.53	28.19	-	-
April	361.5	6.5	27.25	77.43	237.5
Mean	203.17	6.12	28.09	79.99	250.28

(13) Conclusions and Recommendations

SUMMARY, CONCLUSIONS AND RECCOMENDATIONS

Summary

This study was focused on phenological characterization of dragon fruits grown under Capiz State University-Pontevedra condition. Specifically, the study was conducted to answer the following objectives: (1) document the growth stages of dragon fruits grown under Capiz State University-Pontevedra condition in terms of: vegetative bud development, shoot development, development of vegetatively propagated part, reproductive development, flowering, fruit development and fruit maturation; (2) evaluate the cladode traits in terms of: length of the stem, length of areoles, cladode width, distance between areoles, and number of spines; (3) characterize the fruits in terms of: onset of flowering, harvesting of fruits, days from flowering to harvesting, number of bracts, longitudinal diameter, fruit length, weight of the whole fruit, pulp weight, skin weight; pulp yield (g plant^{-1}), and skin thickness; and (4) describe the effect of climate on cladode traits of dragon fruits in terms of stem length, length of areoles, width of the cladode, distance between areoles, and number of spines.

The field experiment was laid out in a Randomized Complete Block Design (RCBD) with four replications. The pre-existing dragon fruit plants consisting of 12 rows were divided into 3 rows each. The first three rows, the 4th to 6th rows, 7th to 9th rows, and 10th to 12th rows represented the first, second, third and fourth replications, respectively. The data on the phonological growth stages of dragon fruits were taken from these rows including the cladode and fruiting characteristics.

Findings of the study revealed that: the length of areole, width of cladodes and number of spines were influence by the months of the year but not the length of the stem or the distance between areole. The width of the cladode was significantly influence by the months of the year

where the widest noticed in February, March, and April with means of 5.41 cm, 5.41 cm, and 5.43 cm, respectively, but comparable from October 2021 to January 2022 with means ranging from 5.28 cm to 5.37 cm. Similarly, the plant cladode width in July and August, with means ranging from 5.15 cm to 5.22 cm and was the smallest in May and June, with means of 5.06 cm and 5.08 cm, respectively and an average ranging from 5.15 cm to 5.32 cm, was comparable to that of those plants from July to November.

On the dragon fruit characters, the distinctive characteristics of the dragon fruit grown at the Capiz State University-Pontevedra Campus includes the start of flowering, fruit harvesting, number of bracts, longitudinal diameter, fruit length, weight of the whole fruit, weight of the pulp, yield of the pulp, weight of the skin, and thickness.

While other types bloom in August or September, it was found that the dragon fruits grown on the Pontevedra campus started to bloom in July and August of 2021. It was discovered that fruit was collected around August or September. It was shown that for plants whose flowering started in July and lasted between 59 and 61 days, the dragon fruits could be collected on average in 60 days. Plants that started flowering in August needed an average of 31 days—a time that ranged from 29 to 33 days—to develop their fruits.

Each dragon fruit had an average diameter of 24.98 cm and had 27 bracts. The average size of the fruits is 11.11 cm, and they weigh 281.36 g on average as a whole. The average pulp yield was found to be 73.44%, and the average pulp weight, which ranges from 50 g to 300 g from lightest to heaviest, was found to be 206.64 g.

Conclusions

Based on the result obtained from the data, the following conclusions were drawn:

1. The length of the stem and the distance between areoles were consistent over all months of the year in the cladode characteristics of dragon fruits. From September 2021 to April 2022, however, dragon fruit developed larger cladodes and longer areoles. Between November 2021 and April 2022, plants developed the most spines.

2. Dragon fruit began to flower in CapSU conditions in either July or August, and fruit harvesting started in August or September. Typically, it takes 29 to 33 days to harvest a dragon fruit, as opposed to 59 to 61 days for other species.

3. The average number of bracts was 26.73, the longitudinal diameter was 24.98 cm, the length of the fruit was 11.11 cm, the weight of the pulp was 206.64 g, the yield of pulp was 73.44%, the weight of the skin was 75.55 g, and the thickness of the skin was 3 mm.

4. The stem length, areole length, or space between cladodes were unaffected by monthly rainfall. However, at a monthly rainfall of less than 67 mm, plants had wider cladodes and the greatest number of spines. The cladode characteristics of dragon fruits were unaffected by daily solar radiation or dew point. The number of spines, areole spacing, areole length, and stem length were unaffected by relative humidity. On the other hand, plants subjected to relative humidity levels under 60% have wider cladodes. At values less than or equal to 250.28, the dew point had a significant impact on cladode features such as areole length, cladode width, the distance between areoles, and the number of spines.

Recommendations

Based on the result and findings of the study, the following are recommended:

1. If the college's dragon fruit project is ever expanded, it should be set up on a new site that is well-drained, ideally with sandy soil that is rich in organic matter, and with enough

funding to implement cultural practices that will promote the crop's better growth and development.

2. Planting should take place to coincide with the crop's reproductive phase in the months that favor the development of longer areoles, a wider cladode, and numerous spines, particularly in December or January.

3. A monthly rainfall of 67 mm must be observed for the growth of a larger cladode and numerous spines.

4. To obtain a broader cladode, a relative humidity of less than 60% must be observed.

5. To achieve longer areoles, a wider cladode, evenly spaced areoles, and the development of many spines, a dew point of less than or equal to 250.28 is preferred.

(14) Output

Production of IEC on the dragon fruit production.

(15) Technology Transfer Activity

Conducted Conduct training on the community for factors affecting phenological stages of the dragon fruit.

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

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